MILITARY SPECIFICATION

TEST SET, RADIO AN/URM-101(*)

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be used to perform a preflight check on Radio Set AN/ARN-21 either on the flight deck of aircraft carriers or on the ground at an air station.

1.2 Classification - The Test Set, Radio AN/URM-101(*) covered by this specification shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size (Inches) W	H	L	Max. Allow. Wt. Lbs	Applicable Paragraph
Test Set, Radio	TS-(*)/URM-101					3.5.1
Antenna	AT-(*)/URM					3.5.2
Attenuator, Fixed	CN-(*)/URM					3.5.3
Cable Assembly, Power, Electrical	CX-3135/U (8'-0")			96		3.5.4
Case, Test Set	CY-(*)/URM-101	10	10	14		3.5.5

NOTE: The parentheses (*), in the type designation throughout this specification, shall be replaced by either a number or letter furnished by the Bureau of Aeronautics upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-C-62	Capacitors, Fixed, Electrolytic (DC, Aluminum, Dry Electrolytic, Polarized)
MIL-C-71	Connectors, "N" for Radio Frequency Cables
MIL-T-945	Test-Equipment, for Use with Electronic Equipment General Specification

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MIL-C-3608	Connectors, "BNC", for Radio Frequency Cables
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-5422	Environmental Testing, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-R-7060	Radio Set AN/ARN-21(*)
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

Bureau of Aeronautics

SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

Military

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

DRAWINGS

Military Standard

MS-91185	Connector, Plug, Electrical (Power, Three-Wire, Grounding Type), Type UP121M
MS-2549	Cable Assembly, Power, Electrical CX-3135/U

OTHER PUBLICATIONS

Joint Army - Navy

NAVSHIPS 900-102	Armed Services Index of R. F. Transmission Lines and Fittings

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2.2 Availability of Documents:

(1) Give both title and number when requesting specifications, standards, publications and drawings. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT

(3) Copies of applicable drawings may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D. C. Attention: Avionics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the Bureau of Aeronautics.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be of the prime consideration. In so doing the following factors shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions as required by Specification MIL-T-945 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall first be approved by the Bureau of Aeronautics.

3.1.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts shall be submitted as non-standard parts only when the material requirements of Specification MIL-E-5400 are not met.

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(2) Non-standard parts of an electrical nature as defined in Specification MIL-E-5400, must be approved before the preproduction model of the equipment is submitted for tests. The request for approval must be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Tube Types - The choice and application of electron tubes used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.2.1 Electron Tube Complement - A tube complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

3.2 Design and Construction

3.2.1 General - The equipment shall fulfill all applicable design and construction requirements of Specification MIL-T-945 unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 25°C (+15°C or -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC

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3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in Specification MIL-T-945 with the following exceptions:

(1) Operation with artic mittens of controls behind front panel doors or covers shall not be required.

(2) The tolerance on the accuracy of the bearing information shall be ±4° and distance information will be ± 2 miles on 5-mile setting and ± 5 miles on 100-mile setting for temperatures below -30°C and above +50°C.

(3) The frequency of the input power shall be 50 to 420 cps.

3.2.4 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.5 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-945 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-T-945 having limits as specified herein. The power required shall not exceed the specified amounts:

AC Power

115 V ± 10%
50 - 420 cps

Single Phase

Power Consumption 150 watts at nominal line voltage

3.2.6 Input Power Connections - All AC input power connections shall utilize AN receptacle AN 3102A-10SL-3P.

3.2.7 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustment shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.7.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.8 *Moisture and Fungus Resistant Treatment* - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.9 Shock - The equipment shall meet the shock requirements of Specification MIL-E-5400.

3.2.10 Nomenclature and Nameplates - Nomenclature, nameplates and marking shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.11 Coaxial Connectors - "N" type connectors shall be in accordance with the requirements of Specification MIL-C-71; and "BNC" type connectors shall be in accordance with the requirements of Specification MIL-C-3608 and Armed Services Index of RF Transmission Lines and Fittings (NAVSHIPS 900-102).

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3.2.12 Interchangeability - The components and accessories of this equipment shall conform to the interchangeability requirements of Specification MIL-T-945.

3.2.13 Total Weight - The total weight of the equipment, including combination case, accessories and handbook shall not exceed 35 pounds.

3.2.14 Cables and Connectors - Cables and connectors shall be in accordance with the requirements of Specification MIL-T-945. Insofar as practical all front panel connectors shall be mounted along the lower edge of the front panel and in no case shall their location be such that the connecting cables will interfere with the operator.

3.2.15 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance - After a warm-up period not to exceed 20 minutes, the test set shall receive the standard TACAN interrogation signal (see 3.5.1.2.11) and radiate a TACAN test signal for airborne equipment as described in 3.5.1.2.10. After storage under standard temperature conditions for 1 hour, the test set shall operate with the required performance after a warm-up period not exceeding 60 seconds.

3.4 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of Specification MIL-T-945.

3.5 Detail Requirements -

3.5.1 Test Set, Radio TS-(*)/ARM-101 - This unit shall meet the following requirements:

3.5.1.1 Function - This unit shall be capable of performing a go-no-go preflight check on Radio Set AN/ARN-21 to determine if it is working correctly.

3.5.1.2 Contents - The equipment shall contain the necessary circuitry to generate a TACAN signal (3.5.1.2.10) when interrogated by the standard TACAN signal (3.5.1.2.11). The test set shall also have a self-contained power supply to operate from the specified power source (3.2.5).

3.5.1.2.1 Output Impedance - The transmitter and receiver shall be connected to a common output connector and shall be designed to operate with a nominal 50 ohm load.

3.5.1.2.2 Power Output - The transmitter power output shall be -30 dbm and shall not vary more than ± 3 db over the range of environmental conditions specified in Specification MIL-T-945 with power set adjustment.

3.5.1.2.3 Sensitivity - The receiver shall have a sensitivity of + 3 dbm and shall not vary more than ± 6 db over the range of environmental conditions specified in Specification MIL-T-945.

3.5.1.2.4 Decoding - The receiver sensitivity shall not deteriorate by more than 3 db for pulse coding within ± 0.25 usec of 12 usec but shall have more than 50% reply count-down when pulse pairs coded more than ± 5 usec from 12 usec are received with a signal level no greater than 30 db above the receiver sensitivity setting.

3.5.1.2.5 Traffic Handling Capacity - The test set shall be capable of receiving standard TACAN interrogations from 4 aircraft simultaneously and replying to at least 70% of all interrogations from each aircraft for distance replies of 5 miles.

3.5.1.2.6 Bearing - The video circuitry shall provide fixed TACAN azimuth information equal to 140° ± 2° (as indicated on an aircraft azimuth indicator.)

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3.5.1.2.7 Distance - The internal delay for distance replies shall correspond to a distance of 5 miles ± 1/2 mile and 100 miles ± 2 miles (as indicated on Indicator, Range ID-310/ARN).

3.5.1.2.8 Identity Tone - The identity tone signal shall consist of a series of paired pulses transmitted at a repetition rate of 1350 pulse pairs per second, phase locked with the tenth harmonic of the 135 cycle reference bearing group. An equalizing pair of pulses shall be transmitted 100 usec ± 20 usec after each identity pair.

3.5.1.2.9 Tacan Test Set Channels - The test set when interrogated by the AN/ARN-21 on channel 3 (1027 Mc) or 18 (1145 Mc) shall reply with a frequency of 964 Mc ± .007% or 1205 Mc ± .007% respectively.

3.5.1.2.10 Characteristics of the Radiated Tacan Test Signal for Airborne Equipment - The airborne Tacan equipment may be tested for go-no-go purposes with a signal having the following characteristics:

(1) Pulse Shape - The pulse envelope as detected by a linear detector shall have a rounded shape.

(2) Pulse Top - The instantaneous amplitude of the pulse shall not fall below a line which is 85% of the maximum voltage amplitude of the pulse and is parallel to the base line.

(3) Rise Time - The time required for the leading edge of the pulse to rise from 10 to 90% of its maximum amplitude shall not be greater than 2.5 usec.

(4) Fall Time - The time required for the trailing edge of the pulse to fall from 90 to 10% of its maximum voltage amplitude shall not be greater than 3 usec.

(5) Pulse Duration - The pulse duration, which is measured between the points on the leading and trailing edges of the pulse which are 50% of the maximum voltage amplitude of the pulse, shall be 6 ± 1 usec.

(6) Pulse Droop - Without amplitude modulation of the composite r-f pulse train, any pulse in a pair, or in the Main Reference Group, shall not deviate from the average peak amplitude by more than ± 10% and any pulse in an Auxiliary Reference Group shall not deviate by more than ± 10%.

(7) Pulse Coding - The pulses shall be coded in pairs with a spacing of 12 ± 1 usec as measured at the 50% maximum voltage amplitude points from the leading edge of the first pulse to the leading edge of the second pulse.

3.5.1.2.10.1 Main Reference Pulse Group - The 15 cycle reference bearing signal, called Auxiliary Reference Group, shall consist of a group of 6 ± 1 pairs of pulses. The spacing between pairs shall be 24 ± 0.8 usec. The repetition rate shall be 135 cps and shall be precisely synchronized with the Main Reference Group; however, the Auxiliary Group which would otherwise coincide in time with the Main Reference Group, shall be removed so that only the Main Reference Group appears in this position in the final output signal.

3.5.1.2.10.2 Identification Test Signal - The identification signal shall consist of a series of parallel pulses transmitted at a repetition rate of 1350 pulse pairs per second, phase-locked within ± 50 usec of the tenth harmonic of the 135 cycle reference bearing signal. The first pulse of an identity pair shall occur 740 ± 50 usec after the first pulse of each Auxiliary Reference Group.

(1) Equalizing Pulse Pair - To preserve a constant duty cycle, an equalizing pair of pulses shall be transmitted 100 ± 20 usec after each identity pair.

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3.5.1.2.10.3 Composite 15 and 135 Cycle Bearing Test Signal - The 15 and 135 cycle bearing signal shall be generated by the composite amplitude modulation of the radio-frequency pulse signals at 15 and 135 cps. The combined amplitude modulation percentage shall be as given in the following formula. The amplitude modulation shall be synchronized with the Main and Auxiliary Reference Group so that a bearing of 140 ° ± 2° will be displayed by a calibrated AN/ARN-21. In the absence of harmonics the envelope of the detected r-f signal shall follow this formula.

$y = 1.0 + A \sin (2\pi f t \pm \theta - \gamma) + B \sin (18\pi f t \pm \phi - 9\gamma)$

where y = normalized composite 15-135 cycle signal amplitude
A represents modulation of 15 cycle component = .18 ± .08
B represents modulation of 135 cycle component = .18 ± .08

ϕ = 10° max. (of 15 cycles)
θ = 10° max. (of 135 cycles)
γ = 140° = bearing to the ground of shipboard station from the point of observation
f = 15.0 cps ± 2%
t = time in seconds from the corrected position of the 10th/pulse of the Main Reference Group. This is computed from the average position of the 12th/ pulse of each Auxiliary Reference Group.

3.5.1.2.10.4 Distance Reply Signal - The distance reply signal shall consist of a pair of pulses transmitted in response to a pair of interrogating pulses. The time delay measured from the leading edge of the second interrogating pulse at the input to the test set to the leading edge of the second reply pulse at the output from the test set shall be 112 ± 6 usec when the test set is adjusted for 5-mile range and 1286 ± 25 usec when the test set is adjusted for 100-mile range.

3.5.1.2.11 Characteristics of Tacan Interrogating Signal - The interrogating signal, radiated by the airborne equipment, shall have the following characteristics:

(1) R-F Pulse Spectrum - The energy level contained in a 0.5 Mc band centered on a frequency ± 0.8 Mc from the channel frequency, shall be at least 23 db below the energy level contained in a 0.5 Mc band centered on the channel frequency. The energy level contained in a 0.5 Mc band centered on a frequency ± 2.0 Mc from the channel frequency, shall be at least 38 db below the energy level contained in a 0.5 Mc band centered on the channel frequency. Each lobe of the spectrum shall be of less amplitude than the adjacent lobe nearer the multiplied oscillator frequency.

(2) CW Output - CW output between pulses shall be at least 80 db below the peak pulse level.

(3) Pulse Shape - The pulse envelope as detected by a linear detector shall have a rounded shape falling within the following limits:

(a) Pulse Top - The instantaneous amplitude of the pulse shall not fall below a line which is 95% of the maximum voltage amplitude of the pulse and is parallel to the base line.

(b) Pulse Rise Time - 2.5 ± 0.5 usec

(c) Pulse Fall Time - 2.5 ± 0.5 usec

(d) Pulse Duration - 3.5 ± 0.5 usec

(4) Pulse Coding - The pulses shall be coded in pairs with a spacing which shall be 12 ± 0.5 usec as measured at the 50% maximum voltage points from the leading edge of the first pulse to the leading edge of the second pulse.

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(5) Pulse Repetition Rate - In the search condition, the interrogating signal from one aircraft shall not exceed 150 pulse pairs per second, however, an interrogation rate in the repair of 135 pairs per second shall be avoided. In the track position, this number shall not exceed 30 pulse pairs per second.

(6) Pulse Repetition Rate Variation - The spacing between successive interrogations shall be sufficiently nonuniform as to preclude one aircraft from locking on to range replies intended for any other equipment interrogating the same beacon.

3. 5. 1. 3 Front Panel Layout - All operational controls, connectors adjustments, and indicators shall be located on the front panel. Prior to fabrication of the first equipment, the contractor shall submit a drawing of the proposed panel layout, including identifying titles and calibration marks, to the Bureau of Aeronautics for approval.

3. 5. 1. 3. 1 Controls - The following controls, labeled as capitalized shall be provided on the front panel of the test set:

(1) POWER ON - A two-position toggle switch which applies AC power to the equipment in the ON position.

(2) RANGE: 5 MILES; 100 MILES - A two-position toggle switch which selects either the fixed 5-mile delay or the 100-mile delay.

(3) POWER SET - A knob which adjusts the power output level as viewed on meter mounted on the front panel.

3. 5. 1. 3. 2 Control Design - All controls, both manual and screw-driver adjusted, shall be located and labeled to provide for ease of operation and maintenance. All controls designed for continuous adjustment shall give smooth electrical performance over the entire range of adjustment and shall have no "critical" points at which they will require special care in setting.

3. 5. 1. 3. 3 Electrical Connections - The following front panel connections, labeled as capitalized, shall be provided:

(1) POWER INPUT; 115 V AC, 50-420 CPS - An AN 3102A-10SL-3P receptacle shall be provided for 115 volts, 50-420 cps power input. Pin B of this receptacle shall be grounded to the chassis.

(2) ANTENNA - A type UG-1052/U receptacle shall be provided as the common output connection to the antenna for the test set transmitter and receiver.

3. 5. 1. 4 Marking - The word "caution" shall be printed in red 1-inch high letters on 2 visible sides of the set. Under the "caution" shall be printed the statement:

"Do not operate on external antenna when antenna separation is less than 20 feet."

3. 5. 1. 5 Standby Heater - The standby heater requirements of Specification MIL-T-945 shall not apply.

3. 5. 1. 6 Blower - The use of a blower will be acceptable provided the contractor can demonstrate the forced air cooling of the equipment is necessary. In case a blower is found to be necessary, the smallest practical size unit shall be used.

3. 5. 2 Antenna AT-(*)/URM - The antenna shall radiate the signal generated by the test set for the airborne equipment. The antenna shall meet all the performance and operational requirements for the test set. The antenna shall be connected to a type "N" connector (IPC-6875 or equal). A type UG-185/U connector shall be mounted on the type "N" connector to provide a

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connector for a 30 db attenuator to allow direct connection to the Tacan equipment under test.

3.5.3 Attenuator, Fixed CN-(*)/URM - A 30 db attenuator shall be supplied as part of the test set to provide a 30 db loss when the Tacan equipment is connected directly to the UG-185/U.

3.5.4 Cable Assembly, Power, Electrical CX-3135/U (8'-0") - The equipment shall be furnished with one 8-foot power cable terminated at one end with a connector capable of mating with a type AN 3102-10SL-3P connector. The other end shall be terminated with a type UP-121M bladed power connector (Drawing MS-91185).

3.5.5 Case, Test Set CY-(*)/URM-101 - The combination case shall conform to Specification MIL-T-945 and shall contain provision for stowing the instruction book, antenna, attenuator and cables in the cover. The combination case shall be watertight in accordance with the requirements of MIL-STD-108.

3.5.5.1 Dimensions - The dimensions of the case shall not exceed 10 inches high by 10 inches wide by 14 inches deep.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4..2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-945. The design and test data shall include the data obtained by the contractor in testing to Specifications MIL-T-945 and MIL-E-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-945 except shock tests which shall be in accordance with Specification MIL-E-5422.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

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4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (90 minutes cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.4 Test Procedures - The procedures used for conducting preproduction tests, and acceptance tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.5 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.6 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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